PROCHLORPERAZINE

Vertigo & nausea

Extrapyramidal effects

CHEMICAL NAME 2-chloro-10-[3-(4-methylpiperazin-1-yl)propyl]-10H-phenothiazine (maleate or mesilate)

CLASS Phenothiazine antiemetic (D2 antagonist)

TRADE NAMES Stemetil · Compazine · Buccastem (buccal)

THERAPEUTIC USES / INDICATIONS

- Severe nausea and vomiting.
- Vertigo and labyrinthine disorders (Ménière's).
- Migraine-associated nausea.
- Schizophrenia and anxiety (higher doses).

ADMINISTRATION

- **PO** tablets; **buccal** tablet placed between upper gum and lip to dissolve.
- **IM** deep injection (IV not recommended in some countries).

DOSAGES

- **Nausea (PO):** 5–10 mg 2–3 times daily (acute 20 mg).
- **Buccal:** 3–6 mg twice daily.
- **IM:** 12.5 mg, then oral after 6 h.
- **Vertigo:** 5 mg three times daily.
- **Children (>10 kg):** 0.25 mg/kg 2–3 times daily (limited use).

PHARMACOKINETICS

- **Onset:** PO 30–40 min; IM 10–20 min. **Duration:** 3–4 h (up to 12 h).
- Metabolised in liver.
- Excreted in urine and bile. **t½:** 6–9 h.

PHARMACODYNAMICS

- Blocks **D2 receptors** in CTZ; antihistaminic and anticholinergic effects aid vestibular nausea.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Drowsiness, dizziness.
- Dry mouth, blurred vision.
- Postural hypotension.

ADVERSE EFFECTS (SERIOUS)

- Acute dystonia, akathisia, parkinsonism (esp. children, young adults).
- Neuroleptic malignant syndrome.
- QT prolongation; agranulocytosis; cholestatic jaundice (rare).
- Increased mortality in elderly with dementia.

CONTRAINDICATIONS & CAUTIONS

- CNS depression, coma.
- Bone-marrow suppression.
- Children <10 kg (dystonia); phaeochromocytoma.
- **Caution:** Parkinson's, epilepsy, glaucoma, elderly.

KEY INTERACTIONS

- **Alcohol, CNS depressants:** ↑ sedation.
- **Levodopa:** antagonism.
- **QT drugs:** caution.

NURSING MANAGEMENT

- Monitor for abnormal movements and BP (postural).
- Fall precautions.
- Assess vertigo/nausea response.

PATIENT EDUCATION

- May cause drowsiness; avoid driving and alcohol.
- Report muscle spasms, restlessness, fever with stiffness.
- Rise slowly.

KEY POINTS

- Phenothiazine antiemetic.
- Extrapyramidal side effects.
- Useful for vertigo-related nausea.
- Buccal form for vomiting patients.